

BER Performance Comparison of Classical and AI-Based BPSK Detection over Time-Varying Rayleigh Fading Channels with Doppler Effect

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Abstract

Reliable signal detection is a fundamental problem in wireless communication systems because fading, noise, and mobility can cause transmitted bits to be detected incorrectly. This project studies the bit error rate (BER) performance of Binary Phase Shift Keying (BPSK) transmission over time-varying Rayleigh fading channels with Doppler effect. First, a classical coherent detector with perfect channel state information (CSI) is implemented as the baseline. Its BER performance is analyzed through Python-based Monte Carlo simulation under multiple Doppler frequencies representing different mobility conditions. Next, an AI-based detector is introduced using a Long Short-Term Memory (LSTM) network trained on time-varying channel sequences. The performance of the classical detector and the LSTM detector is then compared over a range of E_b/N_0 values and Doppler frequencies. The results show that BER generally decreases as E_b/N_0 increases, while Doppler-induced fading affects the channel behavior under slow and fast mobility cases. The LSTM-based detector successfully operates on time-varying channel data, but it does not consistently outperform the classical detector in the current setup. This is reasonable because the classical detector uses perfect CSI and remains highly effective under ideal receiver assumptions. The project shows that LSTM-based detection is feasible for sequential fading channels, but its advantage is more likely to appear under imperfect or delayed channel knowledge.

Keywords— BPSK, Rayleigh fading, Doppler effect, BER, coherent detection, LSTM, machine learning, wireless communications

1. Introduction

Wireless communication systems operate in environments where transmitted signals are affected by noise, attenuation, multipath propagation, and mobility. Unlike wired channels, wireless channels vary over time and space, which makes reliable detection of transmitted information more challenging. One of the most common ways to evaluate the reliability of a digital communication system is through the bit error rate (BER), which measures the probability that a transmitted bit is detected incorrectly.

Among digital modulation methods, Binary Phase Shift Keying (BPSK) is one of the simplest and most widely studied schemes. Because of its simplicity, BPSK is frequently used in theoretical analysis and simulation of communication systems. In Rayleigh fading channels, the received signal amplitude changes randomly due to multipath propagation. When mobility is introduced, the channel also becomes time-varying, and its correlation changes according to the Doppler frequency. This makes signal detection more realistic and more challenging than the static fading case.

This project focuses on the performance of BPSK over time-varying Rayleigh fading channels with Doppler effect. The first goal is to implement a classical coherent detector using perfect channel state information. The second goal is to evaluate BER performance under different Doppler frequencies corresponding to different mobility levels. The third goal is to build an AI-based detector using an LSTM model that can process sequential time-varying channel observations. Finally, the performance of the classical detector and the LSTM detector is compared under Doppler-induced fading conditions.

The motivation of this work is to understand whether a time-aware AI model such as LSTM can provide useful performance under time-varying fading channels. Since wireless channels are naturally sequential, LSTM is a more suitable model than a simple feedforward neural network when time correlation is present.

2. Problem Definition and Motivation

The problem studied in this project is the detection of BPSK symbols transmitted over a time-varying Rayleigh fading wireless channel in the presence of additive white Gaussian noise (AWGN). The receiver must determine whether the transmitted bit was 0 or 1 based on the received signal. The performance metric used is BER.

This problem is important because practical wireless systems often involve user mobility, which causes the channel to vary over time. The rate of channel variation is related to the Doppler frequency. Low Doppler frequency corresponds to slow fading and low mobility, while high Doppler frequency corresponds to fast fading and higher mobility. As the channel varies more rapidly, reliable detection becomes more difficult, especially if channel knowledge is imperfect or delayed.

The project uses a classical coherent detector as the baseline because it represents a standard analytical and practical solution. Then, an LSTM-based detector is used as the AI-based enhancement because LSTM is designed to learn temporal dependencies and is therefore better suited for sequential channel behavior than a simple feedforward model.

3. Theoretical Background

3.1 BPSK Modulation

Binary Phase Shift Keying is a binary digital modulation technique in which each bit is mapped to one of two possible signal values. In this project, the mapping is:

$$\mathbf{b} \in \{0,1\}$$
$$\mathbf{x} = 2\mathbf{b} - 1$$

Thus, 0 maps to -1 and 1 maps to +1. This representation allows simple threshold-based detection at the receiver.

3.2 Time-Varying Rayleigh Fading Channel

In a Rayleigh fading channel, the channel coefficient is modeled as a complex Gaussian random variable:

$$\mathbf{h} \sim \mathbf{CN}(0,1)$$

The received signal is given by:

$$\mathbf{y} = \mathbf{h} \mathbf{x} + \mathbf{n}$$

where $\mathbf{x}$ is the transmitted BPSK symbol, $\mathbf{h}$ is the Rayleigh fading channel coefficient, and $\mathbf{n}$ is complex AWGN. In the time-varying case, the fading coefficient changes over time and is

correlated across symbols. The amount of time variation is controlled by the Doppler frequency f_d . A higher Doppler frequency means the channel changes more rapidly.

In this project, a simple Jakes-like time-correlation model is used. The Doppler autocorrelation is represented by the zeroth-order Bessel function:

$$R(k) = J_0(2\pi f_d T_s k)$$

where f_d is the Doppler frequency, T_s is the symbol duration, and k is the time lag. This model is used to generate time-correlated fading samples.

3.3 Slow Fading and Fast Fading

Different Doppler frequencies represent different mobility conditions:

$f_d = 1 \text{ Hz} \rightarrow$ slow fading / low mobility

$f_d = 10 \text{ Hz} \rightarrow$ medium mobility

$f_d = 100 \text{ Hz} \rightarrow$ fast fading / higher mobility

These three cases are used in the simulation to study how mobility affects BER performance.

3.4 Coherent Detection with Perfect CSI

If the receiver knows the channel coefficient h , it can equalize the received signal as:

$$\mathbf{y}_{eq} = \mathbf{y} / \mathbf{h}$$

Substituting the received signal model gives:

$$\mathbf{y}_{eq} = \mathbf{x} + \mathbf{n} / \mathbf{h}$$

The decision rule for coherent BPSK detection is: if $\text{Re}(\mathbf{y}_{eq}) > 0$, detect bit 1; otherwise, detect bit 0. This detector is simple and effective under perfect CSI.

3.5 Theoretical BER Reference

For coherent BPSK over an i.i.d. Rayleigh fading channel, the theoretical BER is:

$$P_b = 0.5 \times (1 - \sqrt{\gamma / (1 + \gamma)})$$

where γ is the average SNR in linear scale. This expression is used as a reference for the original static Rayleigh baseline. In the Doppler case, the main emphasis is on simulation-based BER analysis under time-varying fading.

4. System Model

The communication system used in this project consists of the following stages:

- Random bit generation
- BPSK modulation
- Time-varying Rayleigh fading channel with Doppler effect
- AWGN addition
- Equalization using perfect CSI for the classical detector
- Classical coherent detection
- AI-based detection using an LSTM classifier

The assumptions are BPSK modulation, Rayleigh flat fading, complex AWGN, time-correlated fading controlled by Doppler frequency, perfect CSI for the classical detector, and BER as the performance metric. For the AI-based part, the model input is a time sequence of received signal and channel features.

5. Python Simulation of the Classical Baseline

5.1 Static Rayleigh Baseline

The original baseline simulation was implemented in Python using NumPy and Matplotlib. For each SNR value, 100000 random bits were generated, mapped to BPSK symbols, passed through a Rayleigh fading channel, corrupted by complex AWGN, equalized using the channel coefficient, and detected using the sign of the real part of the equalized signal. This process was used to validate the theory for the static Rayleigh fading case and establish a correct initial baseline.

5.2 Doppler-Based Classical Simulation

The extended classical simulation uses a time-correlated Rayleigh fading process controlled by Doppler frequency. The simulation was performed for $f_d = 1$ Hz, $f_d = 10$ Hz, and $f_d = 100$ Hz. For each Doppler case and each E_b/N_0 value from 0 dB to 20 dB in steps of 2 dB, random bits were generated, mapped to BPSK symbols, passed through the time-correlated fading process, corrupted

Parameter	Value
Number of bits per SNR point (classical Doppler)	100000
Eb/N0 range	0 dB to 20 dB
Eb/N0 step size	2 dB
Modulation	BPSK
Channel model	Time-varying Rayleigh fading
Doppler frequencies	1 Hz, 10 Hz, 100 Hz
Noise model	Complex AWGN
Detection type	Coherent detection with perfect CSI
Performance metric	BER

by complex AWGN, equalized using perfect CSI, and detected using coherent detection. BER was then computed.

5.3 Simulation Parameters

5.4 Classical Doppler Results

The classical Doppler simulation showed that BER generally decreases as Eb/N0 increases for all Doppler scenarios. The results also demonstrate that mobility affects the BER behavior under time-varying fading.

Eb/N0 (dB)	fd = 1 Hz	fd = 10 Hz	fd = 100 Hz
0	0.137360	0.142590	0.118220
2	0.096890	0.104430	0.079650
4	0.066280	0.076820	0.050120
6	0.039580	0.054420	0.030800

8	0.022720	0.039020	0.019150
10	0.011520	0.027400	0.011880
12	0.005300	0.018840	0.007280
14	0.002040	0.012110	0.004740
16	0.000780	0.008270	0.002830
18	0.000270	0.005380	0.001970
20	0.000180	0.002950	0.001320

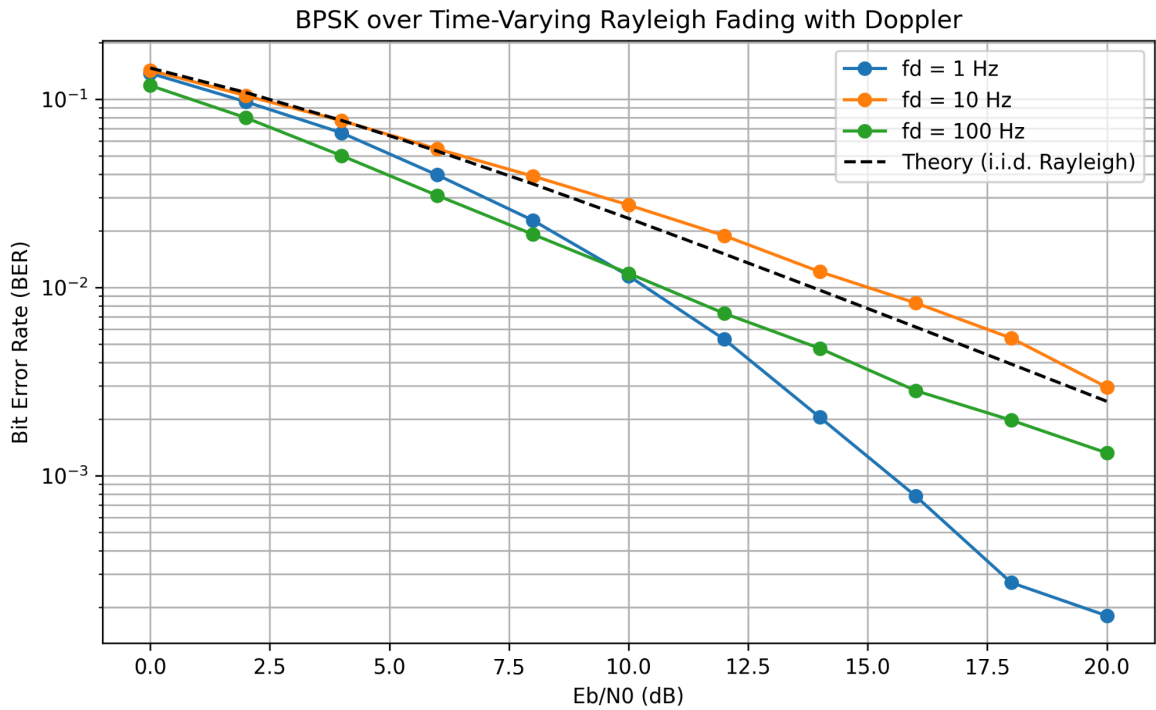


Figure 1. BER performance of classical coherent BPSK detection under time-varying Rayleigh fading with Doppler effect.

6. AI-Based Enhancement with LSTM

6.1 Motivation for LSTM

The original AI baseline in the earlier version of the project used a simple feedforward neural network. However, because the updated channel model is time-varying and sequential, an

LSTM-based detector is more appropriate. LSTM networks are designed to learn temporal structure and sequence dependencies, making them suitable for Doppler-induced fading channels.

6.2 Input Features and Labels

The LSTM detector uses time sequences of features formed from the received signal and channel:

$$\mathbf{X} = [\text{Re}(\mathbf{r}_x), \text{Im}(\mathbf{r}_x), \text{Re}(\mathbf{h}), \text{Im}(\mathbf{h})],$$

where $\mathbf{r}_x$ is the received signal and $\mathbf{h}$ is the fading channel coefficient. The labels are the transmitted bits corresponding to the final time step of each sequence.

6.3 LSTM Model Structure

The AI detector was implemented using PyTorch. The model contains one LSTM layer, a hidden dimension of 32, a fully connected classification head, and a sigmoid output for binary detection. The model processes sequence inputs and predicts the transmitted bit at the last step of the sequence.

6.4 Training and Testing Setup

The LSTM was trained on mixed time-correlated datasets generated from multiple Doppler frequencies and multiple E_b/N_0 values. Training data was formed using continuous channel blocks to preserve sequence structure. The key parameters used were training samples: 100000, testing samples: 100000, sequence length: 8, epochs: 12, batch size: 256, optimizer: Adam, and learning rate: 0.001.

6.5 Performance Metric

As in the classical case, BER was used as the main performance metric:

$$\text{BER} = \text{number of incorrectly detected bits} / \text{total number of bits}.$$

This ensures a fair comparison between the classical detector and the LSTM detector.

7. Results and Discussion

7.1 Static Baseline Reference

The earlier part of the project confirmed that for the static Rayleigh fading case, the simulation closely matches the theoretical BER. It also showed that a simple AI detector based on MLP

performs similarly to the classical detector under ideal conditions with perfect CSI. This static baseline serves as a useful reference before moving to the more realistic Doppler fading case.

7.2 Classical vs LSTM under Doppler Fading

The main updated result compares the classical detector and the LSTM detector under time-varying Rayleigh fading with Doppler effect. The observed BER values are summarized below.

For $f_d = 1$ Hz:

Eb/N0 (dB)	Classical BER	LSTM BER
0	0.169300	0.172162
2	0.110860	0.114328
4	0.039620	0.040753
6	0.012740	0.013501
8	0.040040	0.043743
10	0.022110	0.025272
12	0.005360	0.007581
14	0.000920	0.001560
16	0.002740	0.010731
18	0.002920	0.012521
20	0.000610	0.004180

For $f_d = 10$ Hz:

Eb/N0 (dB)	Classical BER	LSTM BER
0	0.129830	0.130899
2	0.075200	0.076865

4	0.059600	0.061594
6	0.029560	0.030402
8	0.022890	0.026652
10	0.022580	0.028402
12	0.004600	0.006620
14	0.010140	0.021772
16	0.007940	0.021892
18	0.000380	0.003280
20	0.000070	0.001960

For $f_d = 100$ Hz:

Eb/N0 (dB)	Classical BER	LSTM BER
0	0.129090	0.134729
2	0.062070	0.064224
4	0.046230	0.051114
6	0.034820	0.038923
8	0.024340	0.040963
10	0.011610	0.016561
12	0.007600	0.015471
14	0.000040	0.000050
16	0.002500	0.009811
18	0.001820	0.012941
20	0.001050	0.005970

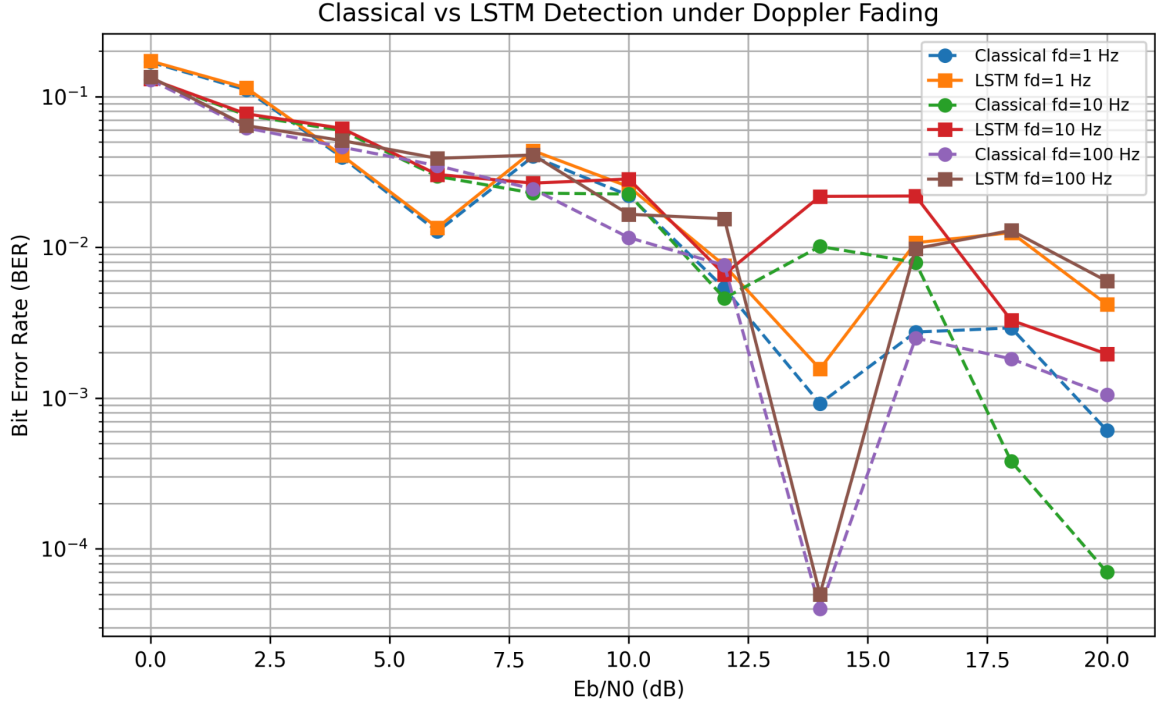


Figure 2. BER comparison of classical coherent detection and LSTM-based detection under Doppler fading.

7.3 Interpretation of Results

Several observations can be made from the Doppler results. First, the BER generally decreases as E_b/N_0 increases for both the classical and LSTM detectors, which is the expected trend. Second, the three Doppler cases represent different channel variation speeds. The time-varying channel influences the BER behavior and makes the scenario more realistic than the original static Rayleigh model. Third, the LSTM detector works successfully on time-sequential fading data, which shows that the AI model can be applied to time-varying wireless channels. However, in the current setup, the LSTM detector does not consistently outperform the classical detector. In most tested cases, the classical detector still achieves lower BER.

7.4 Why the Classical Detector Remains Strong

This result is reasonable because the classical detector uses perfect CSI. When the receiver has ideal knowledge of the channel coefficient, coherent equalization is already a very strong detection method. Therefore, the LSTM detector is competing against a baseline that already has a significant advantage. In this project, the LSTM learns temporal channel behavior, but it still does not

overcome the benefit of perfect CSI available to the classical detector. As a result, the classical method remains more stable and more accurate in most tested scenarios.

7.5 Complexity Discussion

The classical detector is mathematically simple, requires no training data, and has low computational complexity during detection. In contrast, the LSTM detector introduces additional costs such as sequential dataset preparation, model training, parameter tuning, higher computational load, and longer runtime. Since the classical detector still performs better in this setup, the added complexity of the LSTM is not strongly justified under ideal receiver assumptions.

7.6 Practical Insight and Limitations

Although the LSTM detector does not outperform the classical detector here, this does not mean LSTM is not useful in communication systems. Rather, it suggests that its benefits may become clearer under more difficult conditions such as imperfect CSI, delayed CSI, channel estimation error, stronger mobility effects, or more complicated channel models. The main limitations of this project are the classical detector assumes perfect CSI, the time-correlated fading model is simplified, the LSTM model is intentionally modest in size, BER curves still contain some irregularities due to simulation variability, and only BER is compared in detail.

8. Conclusion

This project investigated the BER performance of BPSK over time-varying Rayleigh fading channels with Doppler effect using both a classical coherent detector and an LSTM-based AI detector. The earlier static Rayleigh baseline was first used to validate the correctness of the simulation and establish a reference point. The project was then extended to a more realistic wireless channel by incorporating time-correlated fading controlled by Doppler frequency.

The classical Doppler simulation showed how BER changes under slow and fast fading conditions. An LSTM-based detector was then implemented to handle time-varying channel sequences and compared with the classical detector under the same Doppler scenarios.

The main conclusion is that the LSTM-based detector works successfully under time-varying fading conditions, but it does not consistently outperform the classical detector in the current setup. This is expected because the classical detector has access to perfect CSI and therefore remains highly

effective.

Overall, the project demonstrates that LSTM-based detection is feasible for Doppler fading channels, but its main advantage is likely to appear in more realistic receiver conditions where perfect channel knowledge is not available.

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